# Mina Liu

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## EDUCATION

**Cornell University, College of Engineering** *Ithaca, NY*

B.S. in Computer Science (GPA: 3.9) Expected May 2029

Relevant Coursework: Object-Oriented Programming & Data Structures (Java), Intro to Machine Learning, Functional Programming (OCaml), Discrete Math, Linear Algebra, Multivariable Calculus, Physics I & II

## TECHNICAL SKILLS

- **Languages:** Python, Java, Swift, OCaml, JavaScript, HTML/CSS, Go
- **Libraries & APIs:** NumPy, pandas, Plotly, Gemini API, FAIRINO SDK
- **Infrastructure & Observability:** Docker, Kubernetes, OpenTelemetry, Prometheus, Grafana, Loki, AlloyDB Omni
- **Development & Design:** Git, Xcode, Cursor, Figma, Autodesk Fusion 360

## PROFESSIONAL EXPERIENCE

**Google Database Engineering Collaboration | AlloyDB Omni Observability** *Ithaca, NY*

Software Engineer Intern, Student Program June 2026 – Present

- Build a Python ML pipeline to classify hand-action movements, backed by AlloyDB Omni, Google's PostgreSQL-compatible database; implement reads and writes to persist training data and experiment results for comparison across runs.
- Integrate an observability stack using OpenTelemetry, Prometheus, Loki, and Grafana to collect and visualize ML pipeline metrics and application logs alongside database and infrastructure health, enabling end-to-end monitoring and troubleshooting.
- Develop a Claude-powered diagnostic agent to analyze telemetry across the observability stack, correlate application errors with database and infrastructure signals, and support root-cause analysis of failures within the ML pipeline.

**Robotics Fabrication Research Lab, Cornell University** *Ithaca, NY*

Robotics Researcher, Bowers Undergraduate Research Experience Program June 2026 – Present

- Designed the end-to-end automated knitting workflow and built Python control software with the FAIRINO SDK to coordinate three robotic arms and execute fabrication sequences.
- Implemented coordinate calibration and waypoint generation for a 6 × 6 needle grid, resolving positioning and synchronization failures through iterative debugging and hardware testing.
- Reduced mechanical failures by 20% by designing and 3D-printing custom pinchers, needles, and gears in Fusion 360.

**Human-AI Interaction Design Research Group, Cornell University** *Ithaca, NY*

AI Research Assistant January 2026 – Present

- Developed a Python NLP pipeline in Google Colab to analyze multi-year ChatGPT conversation logs from 8 pilot-study participants, identifying linguistic indicators of cognitive distortions and attachment insecurity.
- Benchmarked Phi-3.5 zero-shot classification, Sentence-BERT (SBERT) embeddings, and rule-based lexicons against manually annotated data using macro-F1 scores
- Built interactive longitudinal visualizations using pandas and Plotly, plotting 7-day slope metrics for rumination, threat language, and negative self-schema to investigate patterns in user well-being over time.

## PROJECTS

**Incubator, Cornell AppDev Project Team** *Ithaca, NY*

Developer & Designer January 2026 – Present

- Building Roomie, a roommate household management iOS app, through rapid prototyping and iterative MVP development, translating a new product concept into a shared noteboard, chore management, and collaborative shopping lists.
- Designed user flows and interactive prototypes in Figma; implemented the iOS frontend in Swift using Cursor.
- Collaborated with backend engineers to integrate APIs and connect frontend features to backend services, supporting shared household data and end-to-end functionality.

**Pac-Man AI (Java)** *Remote*

Developer October 2025 – December 2025

- Developed an autonomous Pac-Man agent using state-based AI, graph representations, and modified Dijkstra shortest-path algorithms to dynamically flee threats, pursue vulnerable ghosts, and collect objectives.
- Implemented custom hash table and binary min-heap data structures, leveraging memoization to optimize pathfinding.